September 3, 2025

Chris Gee
Wexford County Landfill, LLC
990 N Mackinaw Trial
Manton, Michigan 49663

Dear Chris Gee:

SUBJECT: N3862, Wexford County Landfill, Wexford County

On August 14, 2025, the Department of Environment, Great Lakes, and Energy (EGLE), Air Quality Division (AQD) and Materials Management Division (MMD) conducted an announced inspection of Wexford County Landfill, a landfill that is owned and operated by the Green for Life (Company) located at 990 N. Mackinaw Trail, Manton, Michigan. The purpose of this inspection was to evaluate surface methane concentrations at this facility within the requirements of the federal Clean Air Act; Part 55, Air Pollution Control, of the Natural Resources and Environmental Protection Act, 1994 PA 451, as amended (Act 451); the conditions of Renewable Operating Permit (ROP) Number MI-ROP-N3862-2022; National Emission Standards for Hazardous Air Pollutants 40 CFR Part 63 Subpart AAAA (NESHAP); and the Federal Plan requirements for Municipal Solid Waste Landfills that Commenced Construction On or Before July 17, 2014, and have not been modified or reconstructed since July 17, 2014, 40 CFR Part 62 Subpart OOO (Subpart OOO). Compliance with Part 115, Solid Waste Management, of the Natural Resources and Environmental Protection Act, 1994 PA 451, MCL 324.11501 et seq., as amended, and the administrative rules promulgated thereunder (Part 115) was also reviewed.

During the inspection, AQD performed an abbreviated surface emission monitoring (SEM) inspection and found **32** areas with surface methane concentrations greater than 500 ppm above background.

AQD staff used two (2) SEM5000 portable methane detectors. Instrument specifications and calibration records are available in Attachment (1). Detailed spreadsheets/reports of the data collected have already been provided electronically to the Company via email. Attachment (2) provides additional information including an aerial image of the landfill showing the path followed during the inspection and the locations of methane concentrations detected above 500 ppm background. Attachment (3) includes photos taken during the inspection including SEM leaks, and other relevant photos.

The following table shows the results of the SEM inspection conducted during the visit:

AQD ID	LAT	LON	PPM	Description
MK-1	44.35186483	-85.3969235	2534	Penetration
MK-2	44.3512545	-85.39668	658	Sandy bare ground
MK-3	44.35118817	-85.39624483	1186	Penetration
MK-4	44.35123117	-85.39618517	754	Penetration
MK-5	44.35100533	-85.39419417	614	Sandy bare ground
MK-6	44.35103517	-85.394104	1540	Sandy bare ground

MK-7	44.3510745	-85.39381833	909	Sandy bare ground
MK-8	44.351076	-85.39373217	881	Sandy bare ground
MK-9	44.35109417	-85.39362133	1004	Sandy bare ground
MK-10	44.35116867	-85.39334617	873	Sandy bare ground
MK-11	44.35119033	-85.39313733	509	Sandy bare ground
MK-12	44.351347	-85.392842	627	Penetration
MK-13	44.35113167	-85.39222483	743	Penetration Well 49
MK-14	44.35062033	-85.3926985	3895	Penetration
MK-15	44.350159	-85.39299567	510	Penetration Well HC- 4
MK-16	44.34912033	-85.39299617	1369	Penetration
MK-17	44.34883033	-85.393009	1158	Penetration
MK-18	44.34866133	-85.39297817	3585	Penetration GW-37
MK-19	44.35080217	-85.394992	1721	Sandy bare ground
BM-1	44.35121017	-85.396825	2266	Sandy bare ground
BM-2	44.350903	-85.39637167	1250	Penetration
BM-3	44.35098383	-85.39406283	658	Penetration, HC-10
BM-4	44.35097467	-85.39393867	1151	Sandy bare ground
BM-5	44.35095817	-85.39382367	884	Sandy bare ground
BM-6	44.35095367	-85.39369883	932	Sandy bare ground
BM-7	44.35088583	-85.39303567	1141	Penetration, GW-50
BM-8	44.35092433	-85.39293817	584	Sandy bare ground
BM-9	44.3509455	-85.39263033	605	Lose sandy ground
BM-10	44.35085867	-85.3926345	1137	Lose sandy ground
BM-11	44.34922867	-85.3935995	2038	Penetration, GW-41
BM-12	44.348667	-85.393612	580	Penetration
BM-13	44.35101333	-85.39502683	552	Sandy bare ground
Well-1	44.350605	-85.392675	NA	Wellhead pushed over
Well-2	44.348830	-85.392977	NA	Small leachate leak from pipe

Well-3	44.348590	-85.392934	NA	Wellhead pushed over along with locally strong leachate/sewage smell

Note: All methane concentrations above 500 ppm were marked with a red flag. Attachments 1 and 2 provide more detailed information on the SEM survey that was performed. Monitoring was conducted between 8:00 AM and 11:00AM on August 14, 2025.

General SEM Inspection Comments:

This was an initial SEM inspection of the landfill to assess how well the GCCS is working and to identify problem areas.

Note: The Company exceeded the Tier 2 NMOC concentration threshold of 50 Mg/year in April 2016 but failed to submit the required federal GCCS plan till April 2, 2025. The currently installed GCCS will need to comply with applicable SEM requirements of both 40 CFR 63 Subpart AAAA and 40 CFR 62 Subpart OOO once the federal GCCS design plan is approved. Note that the Part 115 Landfill Gas Monitoring and Management Plan was already approved on March 28, 2024, and the Surface Monitoring Plan was approved on April 30, 2024. Therefore, GFL needs to currently be complying with Part 115 and its approved Surface Monitoring Plan which references SEM requirements in 40 CFR Part 60, Subpart XXX.

RECORDS REVIEW

The landfill currently occupies approximately 196.4 acres. Of the 196.4 acres of site property, approximately 137.8 acres are permitted for solid waste disposal. The permitted disposal area is divided into two separate non-contiguous areas commonly designated as the “East Unit” and the “West Unit.” The East Unit, comprising approximately 69.3 acres, includes the original landfill and newer cells, designated as Cell B though Cell J. The West Unit, encompassing 68.5 acres, represents a planned expansion area that is currently undeveloped, with nine individual cells designated Cells 1-9, with Cells 7-9 reserved for Type III Industrial Waste. The landfill originally began receiving refuse in 1977. As of 2018 the overall permitted disposal capacity of the entire facility is 23,260,000 cubic yards (11,519,525 cubic yards for East Unit and 11,740,475 cubic yards for West Unit).

The GCCS consists of 39 vertical wells, 11 horizontal wells, 11 cleanout/leachate related wells, one controlled sump and a 1600 scfm flare. In 2025, installation of 5 horizontal wells, 3 redrilled vertical wells, and 4 vertical wells have already been completed.

Wellhead data was reviewed as part of the SEM. About 30% of the vertical wells show some signs of impairment which is about average compared to other landfills that SEM inspections have been conducted. Most of the impaired wells appeared to be in areas of final cover where there is reduced methane production and likely higher levels of liquid in the wells. In Cell H, there are 6 recently installed horizontal wells that run in a North/South direction across the Cell. These wells have been in operation since this Spring. They collected about 130 scfm of landfill gas as of July 2025. However, according to the latest GCCS map, all of these collectors do not extend south far enough to reach the upper elevations of Cell H where the SEM inspection route passed through. Cell H is under interim cover containing newer waste and is also the lower side slope of the current active cells just to the south.

Records were reviewed to determine if available GCCS system-wide vacuum requirements are being met. Part 115, Section 11512b(2)(e), requires at least 10” water column available at the well furthest from the blower. A check of July 2025 wellhead readings shows all wells have adequate available system pressure. However, numerous well readings were below 10” back in March 2025.

Total average gas flow being collected currently by the landfill and sent to the previous 750 scfm open flare with a blower sized at 600 scfm (30" vacuum) was approximately 630 scfm in December 2024. GFL then removed the old flare and upgraded it to a 1600 scfm (with similar sized blower with 50" vacuum) open flare. Approximately 1100 scfm are currently being collected. A check of the Revised GCCS Design Plan dated April 2, 2025, by GFL, showed a calculated landfill gas generation rate using the LANDGEM model of 1494 scfm for 2024.

The GCCS was reviewed to determine future compliance with Part 115, Section 11512b(2)(i) and (j) requirements. The new 1600 scfm capacity of the flare seems adequate to handle the current amount of collected landfill but there is no backup to it and it will not be large enough for future increases in expected landfill gas emissions. The March 28, 2024, Part 115 Landfill Gas Monitoring and Management Plan for Wexford County Landfill, states that GFL plans to meet the backup requirement by the following:

- Keeping an inventory of critical spare parts for blower repair,
- Access to a back-up company or supplier-owned skid-mounted flare, which can immediately be transported to the site,
- An emergency generator to supply power during utility downtimes OR Rental agreements for emergency generators during power outages.

Liquid level measurement data for the vertical wells was not available for review by the date of this letter. Condensate/leachate is sent to a leachate storage tank prior to deep well injection.

The average H_2S concentration of the landfill gas as measured on 4/09/2024 was 173.3 ppm. At this concentration, offsite odor impacts are possible, especially if a large volume of landfill gas is being emitted or a portion of landfill is emitting a significantly higher H_2S concentration than the landfill average such as from a newer waste cell.

Three (3) quarterly SEM reports were reviewed for 2024 & 2025. The three surveys were conducted by GFL's consultant: Tetra Tech. The results were as follows:

Q4 2024: Five exceedances.

Q1 2025: Two exceedances.

Q2 2025: One exceedance.

Note that no Quarterly SEM exceedances were recorded in Cell H in 2025.

Over the last 12 months, the Company has not logged any odor complaints although there have been odor complaints in recent years, especially north of the landfill. It is thought that the recent change to a higher capacity flare and correspondingly higher levels of gas collected will reduce the potential for more odor complaints going forward. For example, in December 2024, 630 scfm of landfill gas was collected. In August 2025, an average of 1100 scfm was collected.

GFL's Gas Migration Monitoring Plan dated October 27, 2023, was reviewed. Some exceedances have occurred at monitoring Well GP-4R (just East of Existing Cell G-2) and at GP-5R (just East of Existing Cell G-3) but may have been a result of a temporary shutdown of the GCCS at that time.

SEM INSPECTION

The SEM inspection route followed can be viewed from the map in Attachment (2). The route only included areas of interim cover that lacked vegetation. None of the final cover areas which are found on the lower slopes of the east/south and west sides of the landfill were inspected.

Thirty-two **(32)** SEM hits were recorded. This included hits at sandy bare ground areas and wellhead penetrations. (Note: There were several SEM hits in Cell H that were not officially recorded due to excessive amounts of surface methane over a wide area.) On the surface, the wellhead penetrations only appeared to have sand piled around them as the means to seal them.

Cover integrity viewable along the SEM route appeared to be very uniform and intact. However, the cover material being used is sand. The route followed was dry. Erosion rills were completely absent. (Note: Landfill staff indicated that occasionally, there can be significant loss of the sandy cover during high wind events.) Some minor waste flagging was noted in the interim cover area just west of the main active area.

Gas wellheads appeared to be well maintained in good operating condition. However, wellhead ID labels were either completely absence or hard to read. A couple of wellheads were observed to be leaning over with strong leachate odors coming from one of them. (Refer to SEM hit table for GPS coordinates.)

No dust was noted coming from the haul roads. About 1500 tons of MSW are being hauled in by waste hauler trucks per day, which is the highest rate in Northern Michigan. Landfill staff noted that they are currently not accepting any sludges due to odor concerns.

Light to some moderate levels of landfill gas odors were observed upon arrival in the morning downwind of the landfill on N. Mackinaw Trail Road in a rural area. Also, pockets of minor gas odors were noted in the vicinity of SEM hits, including one well with a strong leachate odor. The strong leachate odor emanating from this well, may be due to a near surface condensate pipe leak.

AREAS OF CONCERN

1. Based on review of previous Quarterly SEM surveys and the results of EGLE's SEM inspection, the Company's SEM surveys are not being conducted in a manner thorough enough to properly identify SEM hits or even areas of excessive landfill gas emissions such as what EGLE discovered in Cell H. Also, numerous wellhead penetrations were found to be leaking which are required to be checked on a quarterly basis.

2. Sand is being used as cover material in the interim cover areas. Due to the high permeability of sand, landfill gas will tend to migrate unimpeded up through it unless it is topped with other cover materials. In the upper elevations of the southern portion of Cell H (possibly extending into the northern parts of Cells D & E and G-3), the combination of sand cover, newer waste, lateral gas migration from the current active cells to the south, and lack of gas well collection is resulting in excessive landfill gas emissions. This is most notable in the southeastern part of Cell H (and possibly Cell G-3) which is the part of the cell with the newest waste. It should be noted that the six (6) horizontal collectors that run north to south in Cell H do not extend far enough south to reach the upper elevations of Cell H where the excessive landfill gas emissions were detected. (Refer to Area of Concern on Results Map in Attachment (2)).

3. Distinct and definite offsite landfill gas odors (2 to 3 on the AQD odor scale) were observed by AQD staff despite the recent upgrade to a larger capacity flare and corresponding significant increase in landfill gas collection.

4. Wellhead ID labels were either absent or often illegible when present.

RECOMMENDATIONS

- Address/fix all **32** SEM hits per Part 115 requirements. Particular attention should be given to properly sealing wellhead penetrations. Also address the 3 wellhead issues outlined in the SEM hits table and properly label all wellheads.

- Install vertical wells in Cell H to control the excessive amount of landfill gas emissions as soon as practical especially in the southern portions of the cell near the interface with Cells D & E and G-3.

- Consider using temporary geomembrane liners in all interim cover areas to augment ineffective sand cover.

As a response to this letter, please provide a report to both MMD and AQD that should include the methane exceedances detected by the AQD during this SEM inspection and at a minimum, the results of the required re-monitoring completed pursuant to Part 115 and the actions taken to clear the identified exceedances within 90 days of the date of this letter.

Thank you for your attention to addressing the results of the SEM inspection above and for the cooperation that was extended to us during our inspection of your landfill.

If you have any questions regarding this letter or the actions necessary to address the referenced exceedances, please contact me at the number listed below.

Sincerely,

Mike Kovalchick

Mike Kovalchick
Senior Environmental Engineer
Air Quality Division
517-416-5025

Attachments: three
cc: Brad Myott, EGLE
Scott Miller, EGLE
Shane Nixon, EGLE
Lindsey Wells, EGLE
John Ozoga, EGLE
James Staley, EGLE

Brian Merle, EGLE
Jeff Benya, EGLE
Jill Cellini, EGLE
Tim Unseld, EGLE

<u>Attachment (1):</u>

AQD used two (2) SEM5000 methane detector devices equipped with tunable diode laser absorptionspectroscopy and has GPS location accuracy of 2 to 4 meters. Monitoring was performed on a representative section of the landfill in accordance with EPA Method 21. The instruments were calibrated using calibration gas of zero and 500 ppm of methane. All monitoring and calibration were done between 8:00 AM and 11:00 AM. Monitoring was observed by landfill representatives during the survey.

Weather conditions with upwind and downwind methane concentrations at the start and end of the SEM provided in table below.

Weather Conditions	**Start Time**	**End Time**
Temperature	57° F.	73° F.
Relative Humidity	100%	64%
Wind Speed mph	1 mph	2 mph
Wind Direction	N	Variable
Background methane upwind	**9** ppm	
Background methane downwind	**9** ppm	

As provided in a previous table, thirty-two **(32)** locations were found to have exceeded the 500 ppm above background threshold during the inspection. The attached aerial image of the Wexford County Landfill shows the path followed during the inspection and the locations of methane concentrations above 500 ppm.

ATTACHMENT (2):

An aerial image of the Wexford County landfill and the path followed during the inspection and locations of methane concentrations above 500 ppm. The yellow box shows an Area of Concern.